Case Study 3: Student Performance Trend Analysis

Assignment Case No. 17 | Level: Beginner

01 PROBLEM STATEMENT

Compare assessment scores across multiple examinations, calculate improvements and declines, identify consistently weak subjects, and classify students according to defined criteria.

Problem interpretation: A student's marks across three assessments are stored subject-wise. The program compares the first and last assessment, identifies improvement or decline, and flags subjects with a low final score.

02 OBJECTIVES

Store marks in lists.

Calculate change between assessments.

Identify subjects below a defined mark threshold.

Summarize overall performance.

03 ASSUMPTIONS

Marks are out of 100.

Each subject has marks for three assessments.

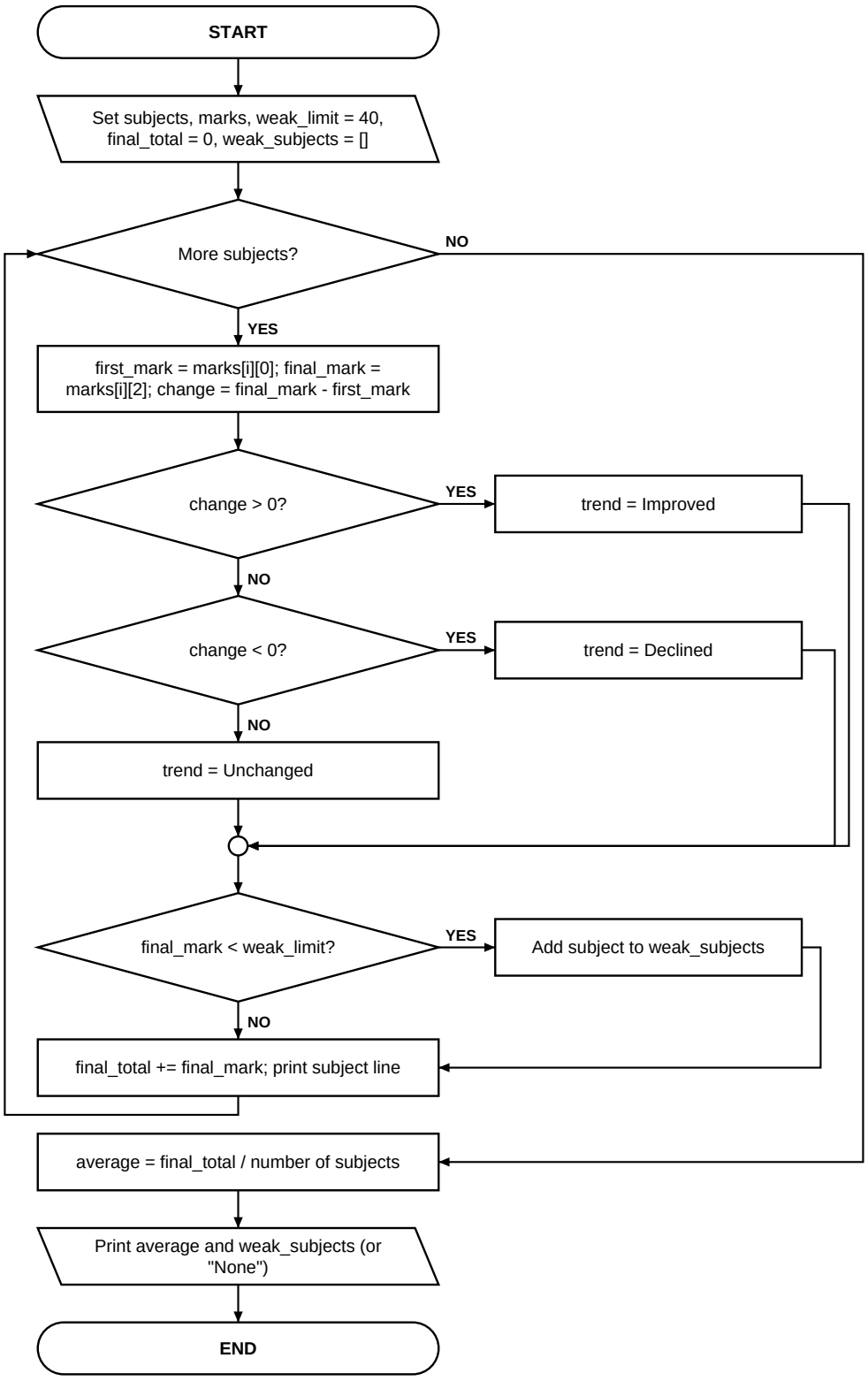
A final mark below 40 is considered weak.

The sample data contains valid marks from 0 to 100.

04 ALGORITHM

1. Start.
2. Store subject names and three assessment scores in lists.
3. For each subject, calculate change from Assessment 1 to Assessment 3.
4. Classify the trend as improved, declined, or unchanged.
5. Flag final marks below 40.
6. Calculate the average final mark.
7. Display the subject-wise analysis and overall average.
8. Stop.

05 FLOWCHART (TEXT FLOW)



06 PSEUDOCODE

```
BEGIN
    subjects = [Maths, English, Python]
    marks = [[55, 60, 72], [70, 68, 65], [35, 38, 39]]
    weak_limit = 40; final_total = 0; weak_subjects = []
    FOR each subject index i
        first_mark = marks[i][0]
        final_mark = marks[i][2]
        change = final_mark - first_mark
        IF change > 0 THEN trend = "Improved"
        ELSE IF change < 0 THEN trend = "Declined"
        ELSE trend = "Unchanged"
        END IF
        IF final_mark < weak_limit THEN add subjects[i] to weak_subjects
        final_total = final_total + final_mark
        PRINT subject, first_mark, final_mark, change, trend
    END FOR
    average = final_total / number of subjects
    PRINT average
    PRINT weak_subjects, or "None" if empty
END
```

07 PYTHON SOURCE CODE

The complete Python program is provided in the separate file 03_Student_Performance_Trend_Analysis_Code.pdf.

08 TEST CASES

INPUT / SCENARIO	CHECK	EXPECTED RESULT
Maths: 55 to 72	Change = +17	Improved
English: 70 to 65	Change = -5	Declined
Python: 35 to 39	Change = +4; final mark 39 is below 40	Improved; flagged as weak subject
Subject with first mark = final mark	Change = 0	Unchanged
Final marks 72, 65, 39	Average = 176 / 3	58.67

09 SAMPLE OUTPUT PREVIEW

```
STUDENT PERFORMANCE TREND
Maths: first=55, final=72, change=+17, Improved
English: first=70, final=65, change=-5, Declined
Python: first=35, final=39, change=+4, Improved
Average final mark: 58.67
Weak subjects: ['Python']
```

10 CONCLUSION

The program uses nested lists to compare assessments, calculate changes, identify weak subjects, and report the average final mark.